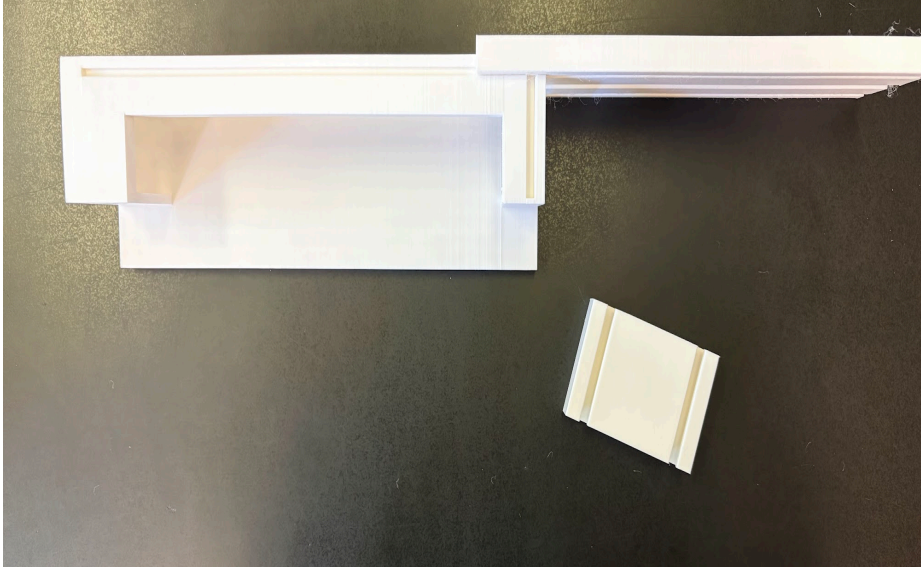


Element D: Design Concept Generation, Analysis, and Selection

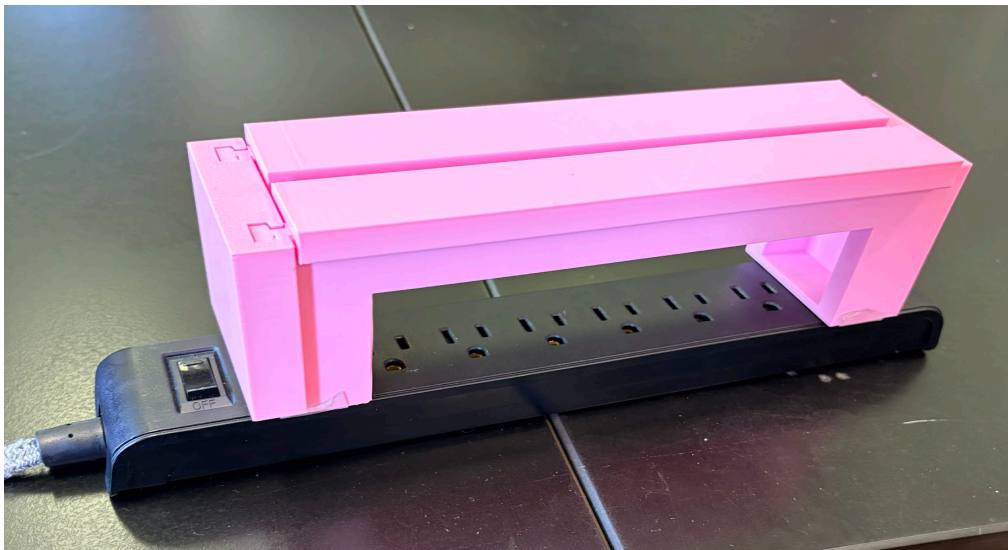
Power cables often come unplugged accidentally for many reasons, like getting kicked or pulled from their socket unintentionally. This affects the general public. The goal of our project is to prevent cables from becoming unplugged by accidentally getting kicked or pulled on. This will help people prevent loss of computer data, loss of aborted creations when 3D printing, or loss of electricity to refrigerators and freezers.

We brainstormed in our notebooks for 10 minutes on our own, then shared our ideas with our team and wrote down any ideas we didn't have. Then we gave feedback on each other's ideas and narrowed them down to five ideas and roughly sketched them. We used a decision matrix to compare our current solution, which is crawling under a table to plug cables back in, to a 3D printed power strip cover, a wall plug clip, velcro, magnets, and a spiral cable. Our decision matrix compared them by how safe they are to use, if it permanently damages outlets, if they cost less than \$.50 per outlet or cable secured, if it was durable enough for at least five installments, if they can be used on a wide variety of outlets and cables, and if they can be installed in under 60 seconds. We weighed safety and cost higher because we thought they were the most important.

(Prototype 1 with sliders partly attached)



(prototype 2 installed on power strip)

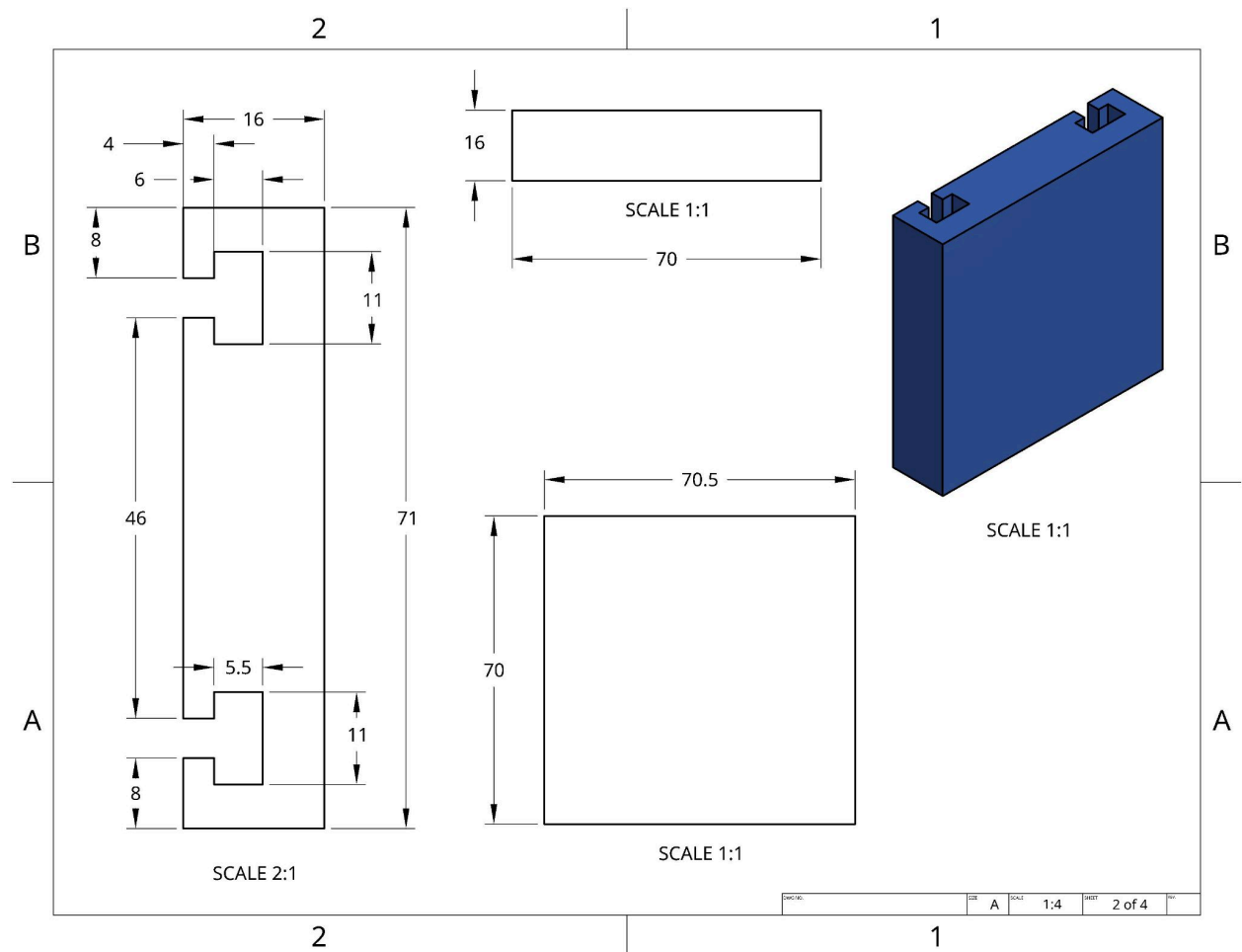


Our final solution is a 3D printed box that goes over a power strip. The box stops the cables from being unplugged. There were other ideas with more points in our design matrix, but we did not include them because it was the most realistic for us to make in class with the available time.

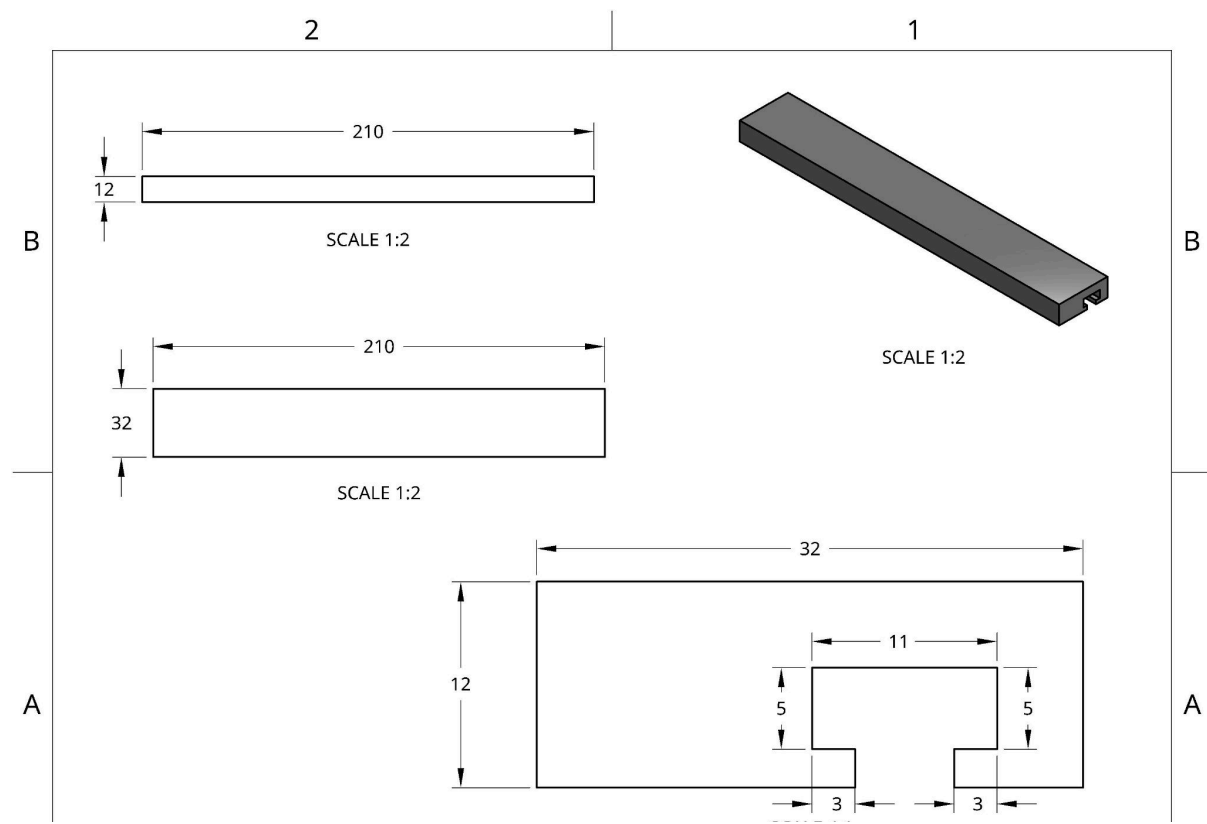
To build our prototype, we started by designing it in Onshape, a CAD modeling

software. We sent the sketch to the 3D printers for printing. Then we assembled the parts we printed and added double-sided tape to the bottom of the cable holder so we could attach it to the power strip. To make this, you would need a 3D printer, PLA (polylactic acid) filament, double-sided adhesive, scissors, and a computer with CAD modeling software. The estimated cost is about five dollars and 50 cents per solution because it takes about a quarter roll of filament, which is 5 dollars and 4 inches of tape, which is about 15 cents.

(Front wall slider)



(top slider)



Technical drawing of a mechanical part, showing multiple views and dimensions. The drawing includes a front view (top left), a side view (top right), a top view (bottom left), and a detail view (bottom right). Dimensions are given in inches.

Front View (Top Left): Shows a rectangular part with a total width of 230 and a total height of 60. The top flange has a thickness of 10. The main body has a height of 40. The bottom flange has a thickness of 30. The part is labeled "SCALE 1:2".

Side View (Top Right): Shows the side profile of the part, indicating a total width of 30 and a total height of 60. The part is labeled "SCALE 1:2".

Top View (Bottom Left): Shows the top surface of the part, indicating a total width of 170 and a total depth of 75. The top flange has a thickness of 10. The main body has a width of 50. The bottom flange has a thickness of 30. The part is labeled "SCALE 1:2".

Detail View (Bottom Right): Shows a detailed view of the corner of the part, indicating a total width of 10 and a total depth of 75. The part is labeled "SCALE 1:1".

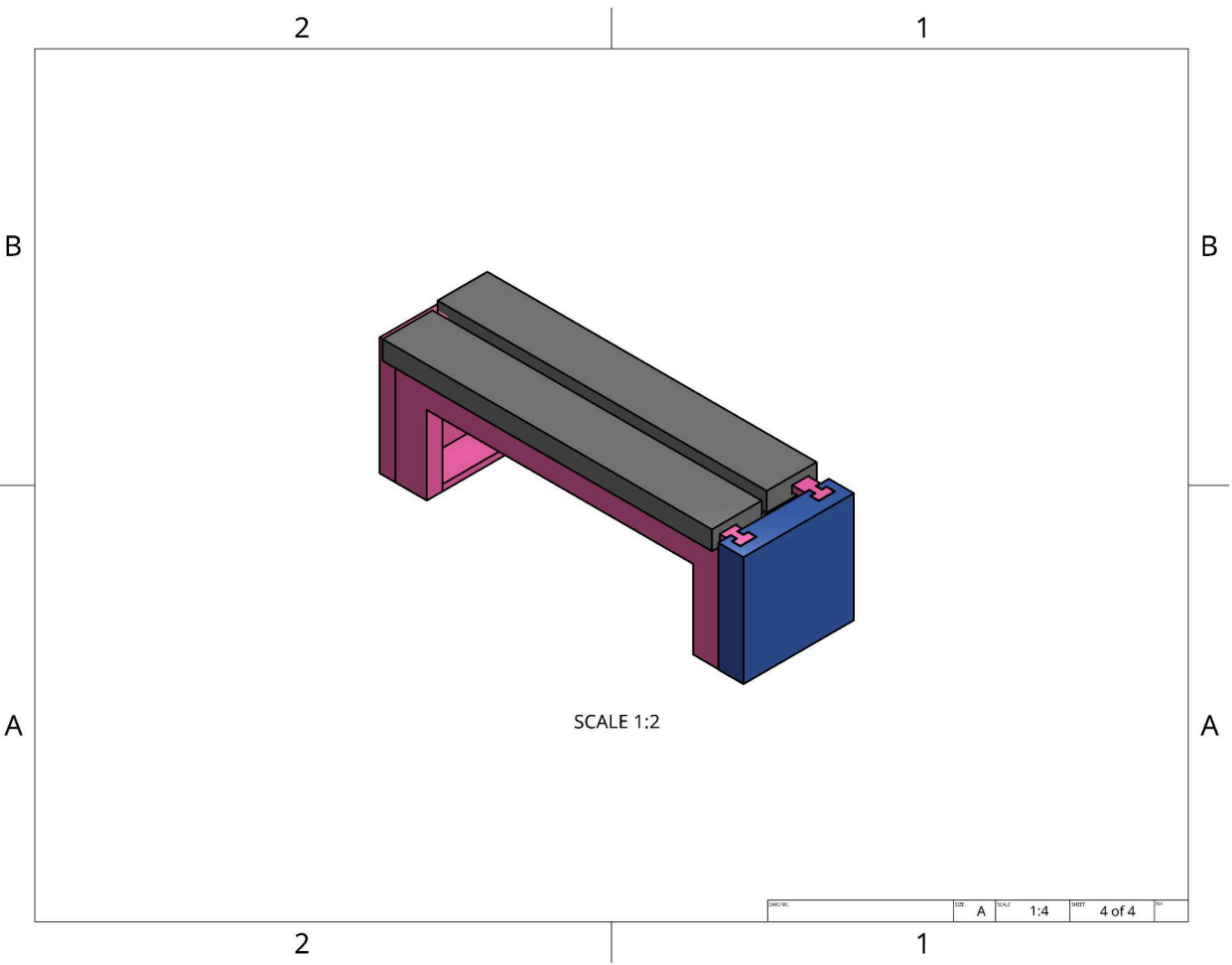
Isometric View (Top Right): Shows a 3D perspective view of the part, indicating a total width of 30 and a total height of 60. The part is labeled "SCALE 1:3".

Section View (Bottom Left): Shows a cross-section of the part, indicating a total width of 30 and a total depth of 75. The part is labeled "SCALE 1:1".

Table:

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES		NAME	DATE
DRAWN	ZACHARY	05/08/2026	
CHECKED			
APPROVED			
DO NOT SCALE DRAWING			
BREAK ALL SHARP EDGES AND REMOVE BURRS			
THIRD ANGLE PROJECTION			
MATERIAL		FINISH	
SIZE		A	
SCALE		1:4	
WEIGHT		1 of 4	

(Full assembly)



Design Requirements	Proposed Solutions					
	Current Solution: Crawling under the table to plug in the cable	Idea #1: 3D printed power strip cover	Idea #2: Wall plug clip	Idea #3: Velcro	Idea #4: Magnets	Idea #5: Spiral Cable
Safe to use-High 2 points	0	+2	+2	+2	-1	+1
Doesn't permanently damage outlets-High 1 points	0	+1	-1	+1	-1	+1
Has to be less than \$0.50 per outlet or cable secured-High 2 points	0	-1	-1	0	-1	-2
Durable enough for at least 5 installments-High 1 points	0	0	+1	-1	+1	+1
Can be used on a large variety of outlets and cables-high 2 point	0	0	-2	+2	+2	+2
Has to be easily implemented, can be installed in under 60 seconds-medium 1 point	0	+1	+1	+1	-1	+1
Has to be easy to use for a child over the age of 10-medium 1 point	0	+1	0	+1	+1	+1
Total Score	0	4	0	7	0	5